

Bhalaji Yadav Kantepalle

Ph.D. Student, Researcher, and Chemical Engineer

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RESEARCH PROFILE

Integrative Life Sciences Ph.D. student and chemical engineer studying stimuli-responsive soft and liquid-crystalline materials through mechanics, quantitative imaging, rheology, and structure-sensitive characterization. Develops reproducible experimental and computational workflows that connect processing, structure, material response, and engineering decisions.

EDUCATION

Ph.D. Student, Integrative Life Sciences

Virginia Commonwealth University

From Jul 2026

Richmond, Virginia

Expected Aug 2026

M.S., Chemical and Life Science Engineering

Virginia Commonwealth University

Richmond, Virginia

- GPA: 3.62
- Thesis: Multi-Criteria Selection of Adhesives for Wearable Textiles
- Advisor: Christina Tang, Ph.D.

Jun 2022

B.Tech., Chemical Engineering

Amrita Vishwa Vidyapeetham

Coimbatore, India

RESEARCH AND TEACHING EXPERIENCE

Graduate Teaching Assistant

Virginia Commonwealth University

Beginning Aug 2026

Richmond, Virginia

- Support undergraduate instruction while beginning doctoral training in the Integrative Life Sciences Ph.D. program.

Sep 2024 - Aug 2026

Graduate Researcher and Metrology Lead

VCU Soft Functional Materials Lab | Advisor: Christina Tang, Ph.D.

Richmond, Virginia

- Develop mechanics-consistent adhesion and fracture-characterization methods for soft, stretchable, and textile-supported interfaces.
- Engineer reproducible Python and OpenCV workflows that synchronize force-displacement data, deformation tracking, and optical colorimetry.
- Apply adhesion testing, tensile analysis, rheology-informed interpretation, optical microscopy, and refractometry to functional soft-material systems.
- Mentored two students in Summer 2026 on characterization planning, NMR-related interpretation, strain-to-color quantification, relationship plots, scientific figures, and poster development.
- Contributed technical writing and regulatory analysis to a phased \$100,000 Commonwealth Cyber Initiative proposal; the \$30,000 prototype and proof-of-concept phase was initially funded, with later funding contingent on satisfactory progress.

PEER-REVIEWED PUBLICATIONS

1. Kantepalle, B. Y.; Epitawala Arachchige, U.; Joung, D.; Tang, C. Multi-Criteria Selection of Adhesives for Wearable Textiles. *Polymers* 2026, 18, 1504. <https://doi.org/10.3390/polym18121504>
2. Caloian, I.; Trapp, J.; Kantepalle, B. Y.; Latimer, P.; Lawton, T. J.; Tang, C. Mechanical Properties of Dual-Layer Electrospun Fiber Mats. *Polymers* 2025, 17, 1777. <https://doi.org/10.3390/polym17131777>

RESEARCH SOFTWARE

Peel Trace Evaluation for Soft Substrates

v1.4.0-rc15

Bhalaji Yadav Kantepalle and Christina Tang

Open-source Jupyter and Colab workflow for protocol-defined analysis of force-displacement peel traces from soft substrates. [Repository](#) | [Archived release](#)

PRESENTATIONS, SEMINAR, AND DEFENSE

1. **M.S. Thesis Defense.** B. Y. Kantepalle, "Multi-Criteria Selection of Adhesives for Wearable Textiles," Virginia Commonwealth University, Richmond, Virginia.

Jul 1, 2026

- Poster Presentation.** B. Y. Kantepalle, U. E. Arachchige, D. Joung, and C. Tang, "Multi-Criteria Selection of Adhesives for Wearable Textiles," PMDC 4th Annual Research Symposium, Virginia Commonwealth University, Richmond, Virginia. Jun 12, 2026
- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," 29th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 22, 2026
- Departmental Research Seminar.** B. Y. Kantepalle, "Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces," Chemical and Life Science Engineering Research Day, Virginia Commonwealth University, Richmond, Virginia. Apr 15, 2026
- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 25, 2026
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," ACS Fall 2025, Washington, DC. Aug 20, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," 28th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 23, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 26, 2025

TEACHING AND SERVICE

Judge, Student Poster Session AIChE Atlantic Regional Conference, Virginia Commonwealth University	Mar 29, 2026
Judge, Chem-E-Car Poster Competition AIChE Mid-Atlantic Regional Conference, Virginia Commonwealth University	Mar 28, 2026
Prototype Demonstrator, Feedback Friday Shelfie Program, VCU da Vinci Center	Mar 20, 2026
Classroom Instructor, Undergraduate Material Balance Virginia Commonwealth University	Nov 11-12, 2025
Student Volunteer Anokha, Amrita University Techfest	Feb 2020

INDUSTRY EXPERIENCE

Technical Project Manager <i>Kreative Organics Pvt. Ltd.</i>	May 2023 - May 2024 Hyderabad, India
- Coordinated quality, engineering, and external stakeholders during a regulated digital-transformation initiative. - Translated process requirements into validation documentation, traceable test protocols, and data-integrity controls. - Standardized chemical nomenclature and analytical workflows to support more consistent technical and business decisions.	
Technical Project Manager Intern <i>Kreative Organics Pvt. Ltd.</i>	Oct 2022 - Apr 2023 Hyderabad, India
Supported digital-transformation, chemical-data standardization, and technical-documentation workflows in regulated manufacturing.	

HONORS AND PROFESSIONAL DEVELOPMENT

- Graduate Assistantship Award, Virginia Commonwealth University, Fall 2025
- Selected Participant, 1st National Neutron Scattering School, Oak Ridge National Laboratory, Sep 2025
- VCU da Vinci Center credentials: Design Thinking, Prototyping, and Pitching and Storytelling, Mar 2026

TECHNICAL SKILLS

Materials and Testing: Adhesion testing, T-peel, fracture mechanics, tensile testing, rheology, viscoelasticity, soft interfaces, wearable textiles

Computational: Python, OpenCV, NumPy, Pandas, SciPy, Matplotlib, image processing, CIE L*a*b* colorimetry, automated analysis

Instrumentation: Universal testing machine, optical microscopy, UV-Vis spectroscopy, Abbe refractometry, 3D printing

Compliance and Process: Validation documentation, data integrity, cGMP, V-model, RCA, CAPA, DOE, Lean manufacturing